

Epilepsy Surgery of the Temporal Lobe in Pediatric Population: A Retrospective Analysis

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BACKGROUND: There is still some reluctance to refer pediatric patients for epilepsy surgery, despite evidence of success in retrospective series.

OBJECTIVE: To describe surgical experience and long-term outcome in pediatric temporal lobe epilepsy (TLE) at a single institution.

METHODS: Retrospective review of pediatric (<18-years-old) TLE patients who underwent surgery between November 1996 and December 2006 at Cleveland Clinic Epilepsy Center. Cox proportional hazard modeling was used to assess outcome predictors.

RESULTS: One hundred thirty pediatric patients met study criteria. Mean time between seizure onset and surgery was 6.3 years. Invasive evaluation was used in 32 patients (24.5%). Hippocampal sclerosis was present in 70 patients (53.8%), either alone or associated in dual pathology. The complication rate was 7%. The seizure-free rates at 1, 2, 5, and 12 years were 76%, 72%, 54%, and 41%, respectively (Kaplan-Meier). With the use of the Engel outcome classification, 98 (75.3%) patients were class I, 11 (8.5%) class II, 9 (7%) class III, and 12 (9.2%) were class IV at last follow-up. Only 4 (3.1%) patients underwent reoperations. Antiepileptic drugs (AEDs) were discontinued in 36 patients (28.3%) in a mean period of 18 months (SD ± 17 months; range, 6-102 months). Although left-sided resection, lower number of preoperative AED trials (≤ 4), and tumor pathology correlated with favorable seizure outcomes, extensive surgical resection remained the only significant outcome predictor after multivariate analysis ($P = .007$; HR = 0.13 [95% confidence interval 0.007-0.64]).

CONCLUSION: Careful selection of surgical candidates by multidisciplinary evaluations is required. Long-term seizure control is achieved successfully with acceptable low complication rates.

KEY WORDS: Epilepsy, Hippocampal sclerosis, Outcome, Pediatric

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Despite the development of multiple antiepileptic drugs (AEDs), 20% to 25% of patients of all ages still have intractable epilepsy, and, in the pediatric population, only 20% of patients have epilepsy of a temporal lobe origin.¹ Temporal lobe surgery in selected patients is one of the most effective means to obtain seizure control.^{2,3}

Reported seizure freedom after surgery in temporal lobe epilepsy (TLE) in the pediatric population ranges from 60% and 86%,⁴⁻⁷ with a tendency

to proceed with early surgical intervention in order to reduce the developmental morbidity of epilepsy and AEDs.^{8,9} Even with the available data, there is some reluctance to refer pediatric patients for epilepsy surgery evaluations, and in many countries the resources to perform such evaluations and procedures are not widespread.¹⁰⁻¹³ Longitudinal analyses of seizure outcome in large pediatric epilepsy cohorts are sparse, as well as the attempts to identify outcome predictors using modern statistical techniques.

In the present study, the surgical experience in a single institution by a single surgeon (W.B.) is analyzed to further understand the role of temporal lobe surgery in this subset of pediatric patients with epilepsy.

ABBREVIATIONS: AED, antiepileptic drug; EEG, electroencephalography; SPECT, single photon emission computed tomography; TLE, temporal lobe epilepsy

METHODS

Between November 1996 and December 2006, 1610 patients with medically intractable epilepsy underwent surgery at The Cleveland Clinic Foundation by the senior author (W.B.); 706 were in the pediatric population (younger than 18 years). Among these patients, 156 had temporal lobe surgery (22.1%). With the approval by our institutional review board, we performed a retrospective analysis in this group, including demographic, clinical, electrophysiological, neuroimaging, surgical, and postoperative data. Patients with previous temporal lobe surgery (12 patients), a pathology report of malignant lesions (1 patient), combined temporal and extratemporal resections (8 patients), and patients lost to follow-up immediately after discharge (5 patients) were excluded from this study.

Preoperative Evaluations

All patients were evaluated by a pediatric epileptologist, and preoperative evaluations were completed in a standardized fashion. Each case was discussed at a weekly Epilepsy Management Conference by neurologists, neurosurgeons, neuroradiologists, neuropsychologists, and a bioethicist. Preoperative evaluations included detailed clinical history, seizure semiology,¹⁴ frequency of seizures,¹⁵ and response to AEDs. Data presented included results of prolonged interictal and ictal video-electroencephalography (EEG) monitoring, magnetic resonance imaging (MRI) with a dedicated epilepsy protocol (T1 coronal volumetric acquisition, T2, and fluid attenuation inversion recovery sequences using 1.5 T/64-MHz MR system), 18-F deoxyglucose positron emission tomography (PET), single photon emission computed tomography (SPECT), and intracarotid amobarbital test.¹⁶ Recommendations of the committee were then discussed with the patients and families, and informed consent was obtained.

Resective Surgical Procedures

The surgical approach was selected according to the preoperative diagnosis, electroencephalographic data, and anatomic characteristics as identified on MRI. The surgical techniques were as follows.

- 1) Corticoamygdalohippocampectomy. This procedure includes removal of a portion of lateral temporal cortex as well as the mesial temporal lobe structures (amygdala, hippocampus, parahippocampus). It was performed in patients with mesial temporal lobe pathology and those with pathology in both the lateral and mesial temporal lobe compartments. The extent of the lateral temporal resection was defined by the senior author (W.B.), with average of 4.5 cm on the dominant side, and 5.5 cm on the nondominant side, with some variation based on cortical vascular anatomy and individualized in lesional cases to include the posterior extent of the lesion. The standard hippocampal en bloc resection included resection of the hippocampus posteriorly to the level of the trigone.
- 2) Lesionectomy plus corticoamygdalohippocampectomy. This extended beyond the corticoamygdalohippocampectomy described above to also include a lesion (benign tumor, focal cortical dysplasia, etc) usually identified on preoperative MRI.
- 3) Lesionectomy only. This procedure included resection of the lesion identified on preoperative imaging (neoplasm or focal cortical dysplasia). Typically, the extent of resection was based on intraoperative findings with or without the use of intraoperative electrocorticography. The mesial temporal lobe structures were not

included in the resection for this group of patients. The choice to perform a lesionectomy vs an extensive corticoamygdalohippocampectomy was individualized by the patient during the patient management conference, and usually relied heavily on concerns for mesial temporal structures epileptogenicity and risk for memory decline should they be resected with the neocortical lesion.

- 4) Tailored temporal neocortical resection and corticoamygdalohippocampectomy. This technique involves mapping of the ictal onset zone and functional temporal lobe cortex before resection. This mapping was usually performed with the use of invasive cortical electrodes (subdural and/or depth electrodes).
- 5) Lateral neocortical resection alone. This included resection of nonlesional temporal lateral neocortex identified as the ictal onset zone by invasive intracranial recordings. Extent of resection was defined by interpretation of ictal and interictal findings during the chronic extraoperative recordings and the location of eloquent temporal cortex.
- 6) Selective amygdalohippocampectomy. This included resection of mesial temporal lobe structures via a transcortical approach through the middle temporal gyrus. It was rarely used in cases where there existed a concern for decline in neuropsychological function following dominant temporal lobe surgery.¹⁷

Invasive recording techniques included stereotactic depth electrode implantation with the use of frameless stereotaxis and scalp-mounted fiducials, craniotomy with implantation of subdural electrode grids (variable configurations), and craniotomy with subdural grid implantation and stereotactic depth electrode insertion by the use of direct visualization of the entry point. Foramen ovale electrode placement was not performed on any of our patients because sphenoidal electrodes were easier and safer to implement.^{18,19}

Postoperative Evaluation

During the immediate postoperative period, all patients received postoperative care, including intensive care unit observation for 1 day, continuing anticonvulsant therapy, and brain MRI on postoperative day 1. Follow-up included visits at 6 weeks and 6 months. The 6-month visit included a high-resolution MRI, neuropsychology, and routine EEG testing. Subsequently, follow-up was obtained at yearly intervals (or more frequently, if indicated) via outpatient visits with the epileptologist or systematic phone follow-up.

Statistical Analysis

Before modeling, the data were summarized with descriptive statistics for each variable including means, medians, and standard deviations for continuous variables and frequencies for categorical variables. For exploratory purposes, an initial analysis of the data was first performed univariably by use of the Wilcoxon rank sum, χ^2 , and Fisher exact tests to compare seizure-free patients with those with seizure recurrence, regardless of follow-up time. This allowed identification of potential prognostic indicators. Variables with a significance level of 10% on initial univariate analysis were then tested in a multivariate Cox proportional hazards regression model. Results were then considered statistically significant at the 5% level. This method allows testing of the correlation of specific variables with outcome while taking into account any interactions and associations among those variables, and their variation with time.

Kaplan-Meier survival analysis was first used to calculate the probability of seizure freedom in the overall group before any outcome predictor

analysis, and later by considering each of the significant risk factors. Statistical significance was tested using the log-rank test and comparison of 95% confidence intervals.

RESULTS

Demographic and Preoperative Evaluation

One hundred thirty patients fulfilled all study criteria and were analyzed; 63 were male (48%). Mean age at surgery was 12.3 years (range, 1-18 years). Handedness was right in 106 patients (81.6%), left in 17 (13%), ambidextrous in 1 (0.8%), and not defined in 6 (4.6%).

Identified predisposing factors for epilepsy, including preoperative MRI findings, are shown in Table 1. The mean time between onset of seizures and surgery was 6.3 years (range, 1-17 years), having between 1 and 10 AED trials before surgical treatment (mean, 4). At least 2 antiepileptic drug trials failed in all patients with chronic epilepsy, with the exception of 17 (13%) patients who underwent surgery after 1 trial because of preoperative imaging features concerning for neoplasm.

Auras were present in 49 patients (37.4%), and the main types of seizures identified were automotor and dialeptic, with 67 (51.1%) and 39 (30%) patients, respectively. History of status epilepticus requiring emergency department admissions was present in 19 patients (14.4%). Seizure frequency according to our institutional classification¹⁵ was daily seizures in 36 patients (27.7%), persistent in 89 (68.4%), and rare in 5 (3.8%). Preoperative psychiatric and behavioral disorders were present in 47 patients (36%).

Electrophysiological interictal and ictal evaluations included sphenoidal electrodes in 96 patients (73.2%). The most frequent

interictal finding was unilateral regional temporal epileptiform abnormalities in 77 patients (59.2%). During the ictal recording, this location was observed in 82 patients (63.0%). Invasive evaluations to localize seizure onset were required in 32 patients (24.5%), 16 (12.2%) with subdural grids only, 9 patients (6.9%) with a combination of subdural grids and depth electrodes, 6 (4.6%) with bilateral amygdala/hippocampus depth electrodes, and 1 (0.7%) with unilateral amygdala/hippocampus depth electrodes. There was a tendency to use invasive evaluations more frequently on left-sided approaches (23 patients, 72.0%), but it did not reach statistical significance ($\chi^2 P = .13$). Additionally, PET was performed in 106 patients (80.9%), ictal SPECT in 32 (24.4%), and intracarotid amobarbital test (Wada test) in 74 (56.4%).

Surgical Procedures

The surgical procedures are detailed in Table 2. Frameless stereotactic guidance was used in 23 patients (17.5%), mainly for lesional resections. The lateral neocortical resection included the superior temporal gyrus and averaged 5.14 cm (SD $\pm$ 0.7; range, 4-8 cm) measured along the middle temporal gyrus on the right side, and 4.23 cm (SD $\pm$ 0.9; range, 3.5-7.5 cm) on the left. Intraoperative electrocorticography was performed in 20 patients (15.2%), while an awake craniotomy was used only in 1 (0.7%). The average estimated blood loss was 196.7 mL (SD $\pm$ 102.0; range, 50-500 mL), while the average surgical time was 4.4 hours (SD $\pm$ 0.97; range, 2.0-7.03 hours). The mean length of stay in hospital for patients not requiring invasive evaluations was 5 days (SD $\pm$ 1.8; range, 3-15 days), while patients with invasive evaluations averaged 12 days (SD $\pm$ 3.06; range, 6-19 days). Postoperative MRI was obtained on day 1 in 72% of patients, and at 6 months in 100%. Analysis of these studies demonstrated complete resection of the lesion in all cases. In cases of corticoamygdalohippocampectomy, and the lateral or basal focal resections, the neuroradiology report described anatomically the structures removed, but no volumetric analysis was obtained (Figures 1 and 2). Overall, hippocampal sclerosis was present in 70 patients (53.8%), either alone or associated with an additional pathology (Table 3). Among the 20 patients with normal MRI

TABLE 1. Predisposing Factors and MRI Findings

	n (%)
Predisposing factors	
Nonidentified	57 (43.8)
Tumor	30 (23.0)
Febrile seizures	19 (14.6)
Perinatal hypoxia	5 (3.8)
Head trauma	5 (3.8)
Tuberous sclerosis	3 (2.3)
Meningitis	2 (1.5)
Hydrocephalus	2 (1.5)
Other (arachnoid cyst, middle cerebral artery remote infarct, cavernoma, neurocysticercosis)	7 (4.9)
Total	130 (100)
Preoperative MRI findings	
Hippocampal volume loss with increased signal on flair	50 (38.4)
Neocortical lesion	44 (33.7)
Normal	20 (15.3)
Amygdala and hippocampal dysmorphia	12 (9.2)
Other (hydrocephalus, arachnoid cyst, heterotopias)	4 (3.0)
Total	130 (100)

TABLE 2. Surgical Procedures^a

Surgical Procedures	N (%)	Individual Rate of Seizure Control, n (%)
CAH	88 (67.7)	48 (53.4)
Lesionectomy plus CAH	18 (13.8)	16 (88.9)
Lesionectomy	18 (13.8)	10 (55.5)
Neocortical resection	3 (2.3)	0 (0)
Tailored temporal lobectomy and AH	2 (1.5)	2 (100)
Selective stereotactic AH	1 (0.7)	1 (100)
	130 (100)	

^aAH, amygdalohippocampectomy; CAH, corticoamygdalohippocampectomy.

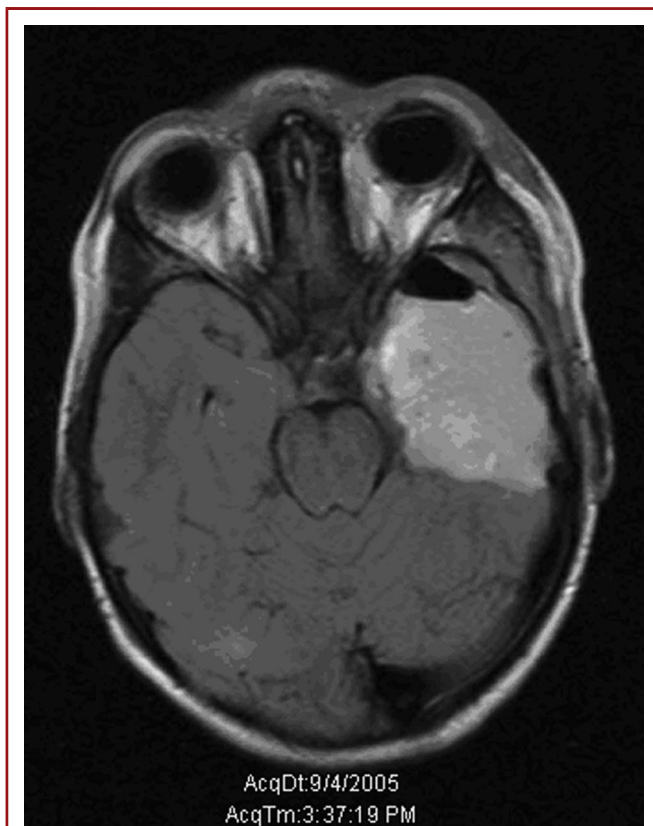


FIGURE 1. Postoperative day 1 MRI (axial flair sequence) example of left corticoamygdalohippocampectomy showing the extent of anterior-posterior lateral temporal and mesial structures resection.

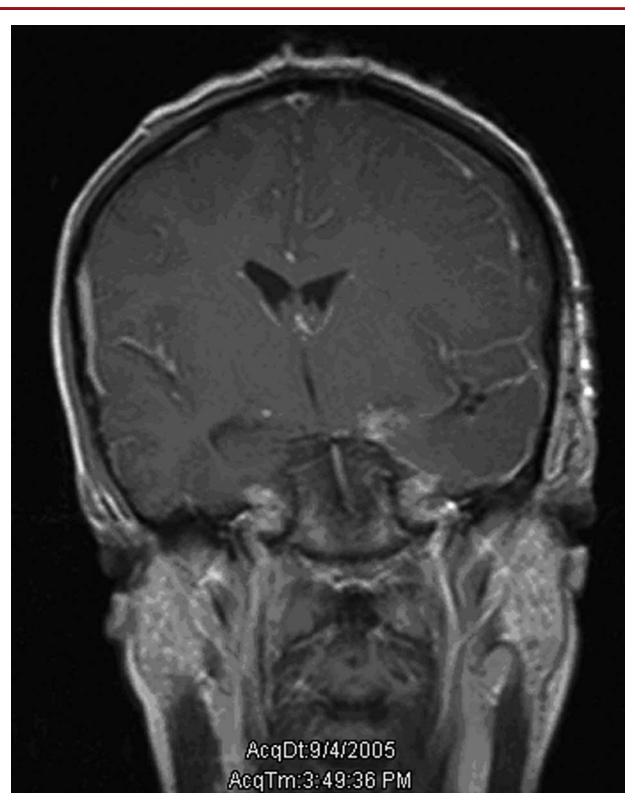


FIGURE 2. Postoperative day 1 MRI (coronal T1 with contrast) example showing margin of subpial dissection, inferior periinsular limit, and upper limit of amygdalohippocampectomy.

results, 9 (40%) had ictal SPECT demonstrating regional temporal-localizing findings, while 16 (72%) underwent the intracarotid sodium amobarbital test (left side only) to document language and memory data. The pathology findings included hippocampal sclerosis in 7 (35%), cortical dysplasia in 5 (25%), chronic inflammation and gliosis in 3 (15%), low-grade glioma in 3 (15%), and dual pathology in 2 (10%).

Outcome

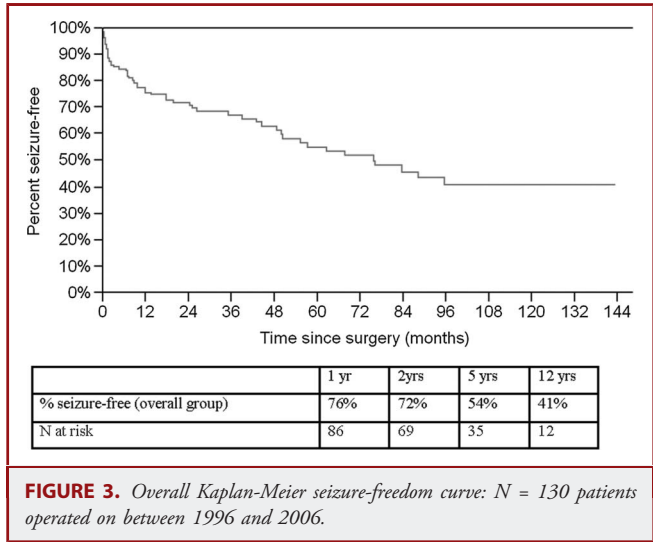
Seizure freedom at 1, 2, 5, and 12 years was 76%, 72%, 54%, and 41%, respectively, as shown in Figure 3 (Kaplan-Meier JMP 8.0). With the use of the Engel outcome classification, 98 (75.3%) patients were class I, 11 (8.5%) class II, 9 (7%) class III, and 12 (9.2%) class IV at last follow-up. Overall, seizures recurred in 54 patients (41%), and seizure control was reobtained in 27 patients (50%) with adjustments of anticonvulsant therapy. The relapse events were associated to AED withdrawal attempts in 18 patients (33.0%), stress and sleep deprivation in 13 (24.4%), and no specific trigger in 23 (42.6%). Only 4 (3.1%) patients with Engel class IV underwent reoperations, and seizure control was achieved in 2 patients (50%)

(Table 4). Complications were present in 9 patients (7%) (detailed in Table 5). Overall, a new diagnosis of depression after temporal lobe surgery was found in 13 patients (10%). Anticonvulsant therapy was successfully discontinued in 36 patients (28.3%) after a mean period of 18 months (SD $\pm$ 17 months; range, 6-102 months). With univariate analysis, side of

TABLE 3. Neuropathology Findings^a

Neuropathology Findings	N (%)	Individual Rate of Seizure Control, n (%)
MTS and FCD	40 (30.7)	22 (55.0)
MTS	30 (23.0)	18 (60.0)
Tumor (ganglioma, low-grade glioma, DNET)	33 (25.2)	24 (72.0)
FCD	14 (10.7)	10 (71.4)
Gliosis and chronic inflammation	8 (6.1)	3 (37.5)
Other	5 (4.3)	1 (20.0)
	130 (100)	

^aMTS, mesial temporal sclerosis; FCD, focal cortical dysplasia; DNET, dysembryoplastic neuroepithelial tumor.



surgery (left), number of AED trials (≤ 4), tumor etiology, and extensive surgical resection (lesionectomy plus corticoamygdalo-hippocampectomy) were predictors for less recurrence (Table 6), whereas multivariate analysis with Cox proportional hazard modeling demonstrated only extensive surgical resection (cortico-amygdalohippocampectomy plus lesionectomy) as significant (Table 7).

DISCUSSION

Patients with epilepsy arising from the temporal lobe are among the best candidates for surgical treatment.³ This option should be considered when seizures persist despite optimal medical treatment. In the adult population, this has been defined as the failure to respond to at least 2 AEDs whether as monotherapies or in combination, with adequate therapeutic levels.²⁰ However, this strategy may not apply to the pediatric population because of the risk of cognitive impairment if seizures persist.^{9,21} Multiple series have reported the success rate of temporal lobe surgery in the pediatric population,^{6,11,22} but, surprisingly, its utilization has not been widespread.¹¹⁻¹³ In our study, the mean time between onset of seizures and surgery was 6.3 years (range, 1-17 years), having between 1 and 10 anticonvulsant trials before surgical treatment (mean, 4).

Optimal selection of surgical candidates requires a thorough evaluation of the clinical, electrophysiological, neuroimaging, and neuropsychology data by a multidisciplinary team including pediatric epileptologists, neuroradiologists, neurosurgeons, neurophysiologists, and neuropsychologists.

The classic clinical presentation of TLE was found with automotor and dialeptic seizures in 51% and 30% of patients, respectively. Different symptomatogenic zones^{14,15} were seen, manifesting as complex motor, hypomotor, and gelastic seizures.

There was an adequate correlation between interictal and ictal EEG for regional temporal localization (60% and 63%,

TABLE 4. Reoperation Characteristics				
Age, y	Preop MRI	Invasive Evaluation	Procedure	Pathology
11	Hippocampal volume loss and increased signal on FLAIR	No	CAH	MTS
18	Normal	No	CAH	MTS/CD
11	Mesial temporal lesion	No	CAH	MTS/CD
15	Hippocampal volume loss and amygdale dysplasia	Yes (Subdural and depth electrodes)	CAH	MTS/CD
			Reoperation	Seizure Control
			Subdural grids followed by redo right temporal neocortical resection	No (Engel IV)
			VNS	No (Engel IV)
			Subdural grids and depth electrodes, followed by redo left temporal neocortical resection	Yes (Engel Ia)
			Subdural grids and depth electrodes, followed by additional cortical parietal resection	Yes (Engel Ia)

^aCAH, corticoamygdalohippocampectomy; CD, cortical dysplasia; FLAIR, fluid attenuation inversion recovery; MTS, mesial temporal sclerosis; Preop, preoperative; Postop, postoperative; VNS, vagal nerve stimulator.

TABLE 5. Overall Complications

Complications	N (%)
Wound infection	2 (1.5)
Visual field deficit	2 (1.5)
Severe brain edema with subdural grids	1 (0.7)
Status epilepticus	1 (0.7)
Subdural/epidural hematoma with subdural grids	1 (0.7)
CSF leak due to external auditory canal injury	1 (0.7)
Transient speech difficulties	1 (0.7)
Total	9 (7.0)

respectively), but multiple locations were also found, including multiregional, bilateral temporal, hemispheric, generalized, and nonlocalizable, as described in previous studies,^{4,5} suggesting wider epileptic spread networks and differential maturation of such networks in the pediatric population.

The classic neuroimaging finding of medial temporal sclerosis (hippocampal volume loss and increased signal on fluid attenuation inversion recovery) was demonstrated in 38.4% of patients. Normal MRI results were seen in only 20 (15%) patients; 12 patients underwent invasive evaluations (54%). Significant outcome differences were not seen in this group of patients. It has been described that patients with normal MRI study results frequently

TABLE 6. Univariate Seizure Outcome Predictors^a

	N	Recurrence, n (%)	P Value
Sex			
Female	63	22 (35)	.14
Male	67	32 (48)	
Side			
Right	51	27 (53)	.03
Left	79	27 (34)	
No. of AED trials			
>4	49	27 (55)	.01
≤4	81	27 (33)	
Tumor			
Present	32	8 (25)	.03
Absent	98	26 (47)	
MRI abnormalities			
Bilateral	11	7 (63)	.13
Other	119	47 (40)	
Normal MRI			
Yes	20	12 (60.0)	.68
No	108	43 (39.0)	
Invasive evaluations			
Yes	32	18 (56.2)	.13
No	98	37 (37.7)	
History of status epilepticus			
Yes	19	9 (47.4)	.41
No	111	46 (41.4)	

^aAED, antiepileptic drug.**TABLE 7. Variables Correlating With Complete Postoperative Seizure Freedom After Applying Multivariate Proportional Hazard Modeling^{a,b}**

Variable	Risk Ratio	95% CI	P Value
Side of surgery (right)	1.59	0.92-2.76	NS
No. of AEDs (>4)	1.46	0.84-2.54	NS
Type of surgery (lesionectomy plus CAH)	0.13	0.007-0.64	.007
Tumor	0.72	0.30-1.48	NS

^aNS, nonsignificant; AED, antiepileptic drug; CAH, corticoamygdalohippocampectomy; CI, confidence interval.^bOverall model log-likelihood P value = .001.

require invasive evaluations because the surgical outcome can be less favorable.²³ Our findings in these patients agree with other studies, suggesting that the lack of an MRI lesion does not necessarily preclude a favorable seizure outcome, mainly in TLE.^{24,25}

Psychopathology was present preoperatively in 47 patients (36%), whereas a new diagnosis of depression was identified in 13 patients (10%). In other series, psychopathology was found in 83%²⁶ because of suggested damage in neural networks.²⁷ Early surgical intervention in pediatric epileptic syndromes may result in more favorable outcomes and prevent irreversible cognitive decline and disability.^{9,28} In the pediatric TLE group, few predictors for neuropsychological outcome after temporal lobectomy exist, and future prospective studies are required.

Language localization studies (Wada) and other functional studies were rarely used because of the pediatric nature of the study population. Functional MRI may preclude the need for invasive evaluations such as the Wada test in older children, given the required understanding of tasks and cooperation for the studies.²⁹

The invasive evaluation was used when discordant noninvasive data, such as EEG or neuroimaging features, were identified, or language localization was necessary for safe surgical treatment (24.5% of series); however, its use was not correlated with seizure outcome after surgery (Table 6). Its use did lead to safe surgical treatment as documented by the low incidence of language dysfunction following surgery. Intraoperative electrocorticography was used less frequently (15%), mainly for tumoral lesions or cortical dysplasias with neocortical onset to help guide surgical resection. This technique is controversial,^{6,30} and it was mainly used to identify and resect the continuous repetitive interictal discharges as described by Palmini et al.³¹

Surgical Procedures and Outcome

Our results indicate that at 1, 2, 5, and 12 years, seizure freedom was reached in 76%, 72%, 54%, and 41%, respectively. It is apparent that, as time passes, the seizure freedom rate declines. Although the reasons for this remain unknown, this decline may reflect secondary epileptogenesis or incomplete resection of the epileptogenic zone. More work is necessary to identify predictors

for seizure recurrence after pediatric temporal lobe surgery. Nonetheless, our results with surgical resection agree with other published series (Table 8).^{5,6,22,32-35} Overall, a favorable outcome was achieved in 84% of patients (Engel class I with 75.5%, and class II with 8.5%), correlating with previous surgical series ranging from 60% to 86%.^{5,6,8,33,36-38} Results of temporal lobe surgery in children are less favorable than in the adult population previously reported at our institution (53% seizure freedom at 10 years),³⁹ which may be related to fewer opportunities to identify the ictal onset zone because of age-related limitations in performing additional testing (ictal SPECT, invasive evaluations, Wada test). It may also reflect different unknown intrinsic pathophysiologic mechanisms accounting for epilepsy recurrence in the 2 populations.

Theoretically, the most important factor for seizure control among the different surgical techniques is the complete resection of the epileptogenic zone.⁴⁰ While preoperative testing and invasive recordings can lead to a hypothesis of the epileptogenic zone, only the seizure outcome after surgery over time will determine whether the area resected was truly the region that generated the epileptic seizures.⁴¹ In our study, extensive resection for lesional epilepsy (lesionectomy plus corticoamygdalohippocampectomy) led to better seizure freedom, whereas, in our nonlesional epilepsy cases, the results were similar to previous descriptions.^{35,39,42,43} Given that the pathology in pediatric temporal lobe epilepsies is often widespread, the role for minimally invasive approaches such as selective stereotactic amygdalohippocampectomy should rarely be used.

Overall, recurrence of seizures was present in 41% of patients, reobtaining satisfactory seizure control in 72% of patients with 50% Engel class I, and 22% Engel class II at last follow-up. Only 4 patients (3.1%) underwent reoperation, with improvement in 2 (Table 4). On multivariate analysis, only extensive surgical resection (corticoamygdalohippocampectomy plus lesionectomy) demonstrated significance. There was some tendency (although no statistical significance) for higher seizure recurrence in nondominant temporal lobe cases. This can be the result of a less strict selection criteria for surgery because of less concern for language deterioration following surgery. Many of these patients had no other options, and surgery perhaps was considered out of compassionate use. Other risk factors such as diffuse pathology, bilateral abnormalities, preoperative secondarily generalized seizures, and febrile seizures did not show statistical significance in our study.^{2,39}

Anticonvulsant therapy was discontinued in 36 patients (28.3%), at a mean time of 18 months after surgery. Discontinuation of AED therapy after surgery is controversial, but it is usually recommended to wait for at least 1 year of seizure freedom and absence of epileptic activity on EEG.^{44,45}

Our overall complication rate was 7%, including wound infections, visual-field deficit, cerebral edema, subdural/epidural hematoma, CSF leak, and speech deficit. There was no mortality or long-term disabling morbidity. In previous series, the complication rates ranged from 2% to 8%.^{6,33,36,46,47} Visual-field assessment was performed by confrontation testing, with the expected superior

TABLE 8. Pediatric Temporal Lobe Epilepsy Surgery Series^a

	Period	n	Age, y	Predominant Surgery Type/%	Follow-up Range, mo	Seizure Freedom, %	Overall Complication Rate, %	Reoperations, %
Sinclair et al, 2003 ²²	1988-2000	42	≤16	N/A	12-144	76.0	4.0	N/A
Clusmann et al, 2004 ³⁵	1989-2000	89	<18	CAH/39.0	14-118	87.0	10.1	1.1
Mittal et al, 2005 ⁶	1985-2000	109	≤18	CAH/52.0	60-240	81.7	5.5	21.0
Matton et al, 2008 ⁵	1979-2003	20	<5	N/A	2-23	65.0	15.0	10.0
Kim et al, 2008 ³³	1993-2005	59	≤18	CAH/79.6	12-168	88.0	47.4	6.7
Kan et al, 2008 ³⁴	1998-2006	33	<21	N/A	12-96	85.0	3.0	6.0
Lee et al, 2010 ³²	2006-2008	19	≤18	CAH/73.7	13-42	63.2	15.7	N/A
Present study, 2011	1996-2006	130	≤18	CAH/67.0	12-144	76.0 for 1 y 72.0 for 2 y 54.0 for 5 y 41.0 for 12 y	7.0	3.1

^aCAH, corticoamygdalohippocampectomy; N/A, not available.

quadrantanopsia the most common finding. This finding was present in 2 patients but did not cause any functional impairment in daily activities. One patient with transient speech deficit recovered entirely with speech therapy and time. Among our complications, 4 patients required reoperation. Two of these patients underwent emergent removal of subdural grids: 1 for severe postoperative brain edema, and 1 for acute subdural/epidural hematoma. The other 2 patients requiring reoperation had postoperative wound infections requiring wound revision and drainage. One of these patients required bone flap removal and 8 weeks of intravenous antibiotics.

Pathology

Our most frequent finding was hippocampal sclerosis, found in 53% of cases. In 23% it was the only pathological finding and in 30% it occurred in combination with cortical dysplasia (dual pathology). Dual pathology is defined as mesial temporal sclerosis together with an additional epileptogenic lesion identified by neuroimaging, or additional microscopic changes revealed only by pathological examination.⁴⁸

In contrast to the adult population, hippocampal sclerosis is less frequent, while dual pathology is more frequent.^{4,49} Both conditions have similar rates of seizure control after surgery, and surgical resection, including the 2 areas of dual pathology, did not affect seizure prognosis in our univariate and multivariate analysis. This is likely because the most common surgical procedure for TLE typically includes both the regions where the dual pathology is localized.

CONCLUSION

The present series in pediatric TLE is one of the largest reported to date, demonstrating efficacy and a low rate of complications. We encourage an early identification of surgical candidacy and treatment in this population to reduce the long-term risk of neurocognitive decline associated with ongoing medically refractory epilepsy. Clearly, more data are necessary over an extended long-term follow-up in this group to better define neurocognitive and quality-of-life outcomes. The surgical technique applied should be individualized by consideration of the preoperative data and suspected pathology. With an experienced surgical team, surgery can be safely performed in young patients. Referral to an experienced surgical center should be considered in all patients for whom 2 trials of anticonvulsant therapy have failed.

Disclosure

The authors have no personal financial or institutional interest in any of the drugs, materials, or devices described in this article.

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COMMENT

This is an interesting report on the impressively large experience with 130 children and adolescents operated on for temporal lobe epilepsy (TLE) by the senior author (W.B.). It is noteworthy that the vast experience of a large epilepsy surgery center is necessary to provide such an overview, because pediatric cases are less frequent than adult cases, and the temporal lobe origin of seizures in pediatric patients is seen in about a fifth of cases, only. Thus, defining and analyzing this probably largest cohort of pediatric patients operated on for TLE contributes a solid and the best basis, at present, for information in this field. There will probably not be any comparable prospective studies in the near future.

Results with respect to seizure control were especially successful for the first years, then a number of seizure relapses was documented (76% seizure freedom initially, down to 41% after 12 years). However, patient numbers decreased with duration of follow-up and a majority of relapses (57%) was triggered. Nevertheless, outcome with Engel classification at last follow-up resulted in 75% of patients assigned to class 1 (seizure-free), and overall 84% had a favorable outcome (class 1 and 2). The clinical relevance of these recurrences is thus debatable and should not be used to prevent surgical treatment in children with TLE. The complication rate was 7%, most of these just minor sequels, without serious permanent morbidity.

It is interesting to read about the outcome predictors: upon multivariate analysis, only more extensive resection reached a level of significance. This is in line with our previously published data and supports the notion that pediatric TLE with seizure onset early during life is probably associated with more widespread pathology than typical adult TLE. The high rate of dual pathology is another fact arguing for more extensive resection in children compared with adults.

The results with 20 MR-negative cases should be regarded with caution, because nearly all patients had clear neuropathological findings, suggesting that, eg, sensitivity in the 1990s was not as high as it is today, especially with high-field (3 Tesla) MR imaging.

This article provides an important and clearly positive risk-benefit ratio in favor of surgery for pediatric patients with TLE. Thus, these data will contribute to a more widespread acceptance of epilepsy surgery for pediatric TLE with resulting good individual chances of seizure control, and better psychosocial development for many patients. It is noteworthy, in this respect, that persisting neuropsychological deficits are a lesser concern compared with adult patients treated for TLE.¹

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